HIDDEN SIGNAL

QuantumTrace

Cryptographic inventory with inspectable evidence



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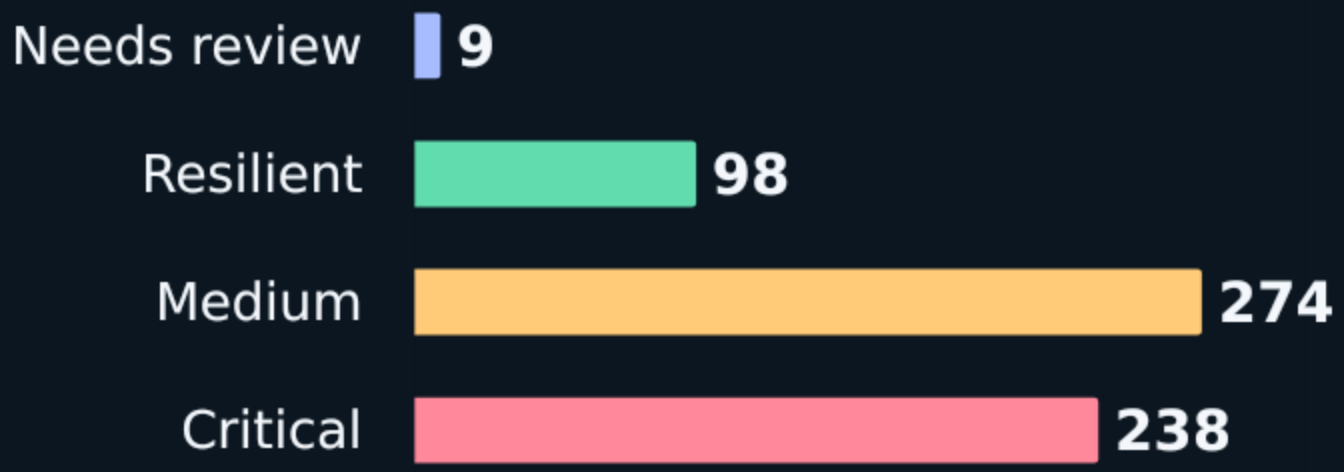
Cryptographic evidence

Input	Detection	Evidence
Python, Java and Go	Supported calls and parameters	Static call
Cloud JSON	Known fields and provider mappings	Configured
TLS captures and JSON	Handshake identifiers and events	Offered or negotiated

One inventory records the source, parameters and assessment rule

Results on the supplied data

1,574 observations = 619 in cards + 955 TLS offers



61.5

Primitive exposure index

610 rated observations

28 analysed files. 7 unsupported metadata files. 0 errors.

Counts describe occurrences. The index is a project heuristic.

New inputs and missing parameters

Example	Key evidence	Challenge tier
01_aes128.py	token_bytes(16)	AES-128: medium
Same file after edit	token_bytes(32)	AES-256: resilient
04_aes_desconhecido.py	AESGCM(chave)	Needs review

8 additional files. 9 observations. Index 37.5 before the edit.

A key-size change updates the inventory. Missing size stays unresolved.

Evidence for migration decisions

86

automated tests passed

Manual expectations across
24 supplied source files

Traceable findings

Evidence, rules and next steps grouped by file or resource

Unresolved cases stay visible

Production accuracy remains unmeasured

Next: broader detection, independent evaluation and CycloneDX

Primitive exposure index

Tier	Weight	Extra cases
Critical	100	2
Medium	50	2
Resilient	0	4
Needs review	Excluded	1

37.5

200 + 100 + 0 = 300

300 / 8 rated observations

Challenge: 37,500 / 610 = 61.5

Unknowns and TLS offers stay outside the mean.

Assessment rules

Primitive	Tier	Basis
RSA, ECDH, ECDSA	Critical	Shor exposure
AES-128 / AES-192	Medium	Challenge policy
AES-256, final PQC families	Resilient	Primitive family
Hashes	Context dependent	Project extension
HMAC and password KDFs	Needs review	Secret / construction review

Hash rules use ideal query models and a project target of 128 bits

The model does not estimate practical attack time or assign NIST categories.

TLS offers and hybrid groups

97 sessions with ServerHello. 7 selected hybrid groups.

Evidence	Inventory status	Interpretation
ClientHello options	Offered	No proof of selection
ServerHello fields	Negotiated	Selected supported mechanism
X25519MLKEM768	Both components	Combined mechanism needs review

A critical classical component does not prove the hybrid is broken